Class: **CAPE Pure Mathematics Unit 1**

Topic: **Past Paper June 2025 – Paper 2 – Solutions**

## June 2025 – Question 1 – Solution

(a) The function $f(x) = px^3 + qx - 6$ has a factor of $x + 2$ and a remainder of $-12$ when divided by $x - 1$.

(i) Show that $p = 1$ and $q = -7$. [8]

$f(x) = px^3 + qx - 6$

Since $f(x)$ has a factor of $(x + 2)$, then by the Factor Theorem, $f(-2) = 0$.

$f(-2) = p(-2)^3 + q(-2) - 6$

$0 = -8p - 2q - 6$

$(\div 2)$

$0 = -4p - q - 3$

$q = -4p - 3$ → Equation 1

Since $f(x)$ has a remainder of $-12$ when divided by $(x - 1)$, then by the Remainder Theorem, $f(1) = -12$.

$f(1) = p(1)^3 + q(1) - 6$

$-12 = p + q - 6$

$-12 + 6 = p + q$

$-6 = p + q$

$-6 - p = q$ → Equation 2

Equating Equation 1 and Equation 2 gives:

$-4p - 3 = -6 - p$

$-4p + p = -6 + 3$

$-3p = -3$

$p = \frac{-3}{-3}$

$p = 1$

Substituting $p = 1$ into Equation 2 gives:

$q = -6 - (1)$

$= -7$

$\therefore$ The values are $p = 1$ and $q = -7$.

Q.E.D.

(ii) Determine the **other** TWO factors of $f$. [5]

$f(x) = px^3 + qx - 6$

Since $p = 1$ and $q = -7$, then we have,

$f(x) = x^3 - 7x - 6$

$= x^3 + 0x^2 - 7x - 6$

We are given that $x + 2$ is a factor of $f(x)$.

Now,

By long division,

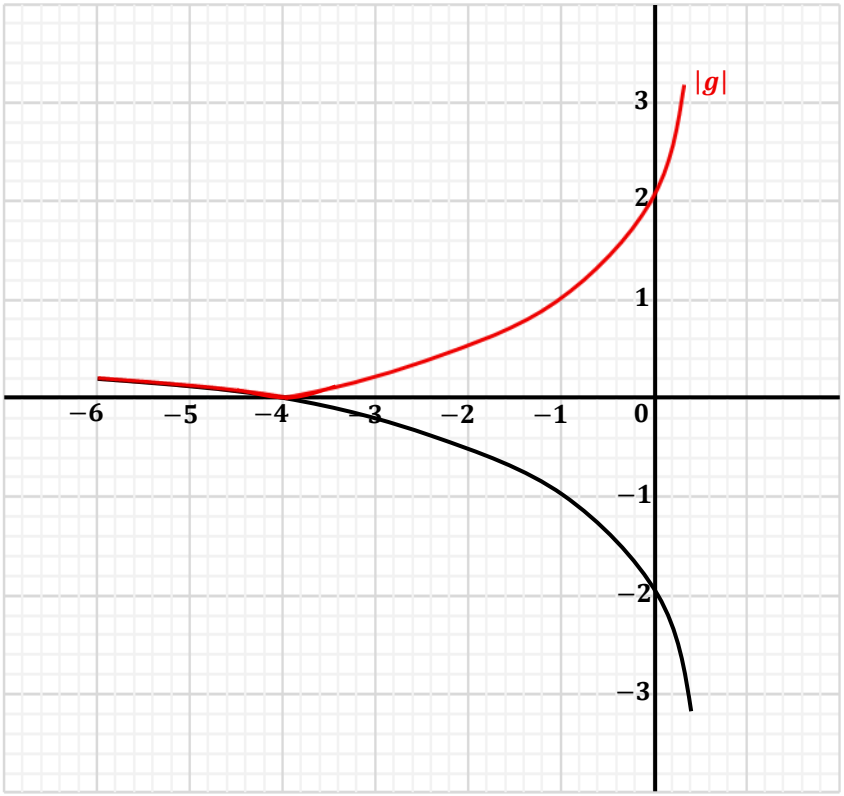
$$
\begin{array}{r|l}
 & x^2 - 2x - 3 \\
\hline
x + 2 & x^3 + 0x^2 - 7x - 6 \\
 & x^3 + 2x^2 \\
\cline{2-2}
 & -2x^2 - 7x \\
 & -2x^2 - 4x \\
\cline{2-2}
 & -3x - 6 \\
 & -3x - 6 \\
\cline{2-2}
 & 0 \\
\cline{2-2}
\end{array}
$$

So, we have,

$$
\begin{aligned}
f(x) &= x^3 - 7x - 6 \\
&= (x + 2)(x^2 - 2x - 3) \\
&= (x + 2)(x + 1)(x - 3)
\end{aligned}
$$

$\therefore$ The other two factors of $f(x)$ are $(x + 1)$ and $(x - 3)$.

(b) The diagram below, **not drawn to scale**, shows the graph of $g(x) = \frac{2x+8}{5x-4}$ for $x \leq 0.5$.




On the same axes, sketch and label the graph of $|g|$. [3]

For the graph of $|g|$, we reflect the part of $g(x)$ that is below the $x$-axis, upward above the $x$-axis. The $y$-values become positive.

(c) Let $f: \mathbb{R} - \{0\} \to \mathbb{R}$ be defined by $f(x) = 1 + \frac{1}{x}$.

(i) By providing a proof or a counterexample, determine whether $f$ is

(a) injective [4]

Since $f(x) = 1 + \frac{1}{x}$,

Then $f(a) = 1 + \frac{1}{a}$ and $f(b) = 1 + \frac{1}{b}$.

Suppose $f(a) = f(b)$.

Then, we have,

$$1 + \frac{1}{a} = 1 + \frac{1}{b}$$

$$\frac{1}{a} = \frac{1}{b}$$

$$a = b$$

Since $f(a) = f(b) \Rightarrow a = b$, then $f(x)$ is injective.

(b) surjective [5]

Let $y \in \mathbb{R}$.

$$y = 1 + \frac{1}{x}$$

$$y - 1 = \frac{1}{x}$$

$$x(y - 1) = 1$$

$$x = \frac{1}{y-1}$$

Consider $y = 1$ where $1 \in \mathbb{R}$.

$x = \frac{1}{(1)-1}$

$= \frac{1}{0} \notin \mathbb{R} - \{0\}$

Since there exists a $y \in \mathbb{R}$ which does not map onto a corresponding $x \in \mathbb{R} - \{0\}$, then $f$ is not surjective.

**Total: 25 marks**

## June 2025 – Question 2 – Solution

(a) (i) Let $b, m$ and $n$ be positive real numbers. Given that $\log_b 3 = m$ and $\log_b 4 = n$,

show that $\log_b \sqrt{\frac{27}{16}} = \frac{3m}{2} - n$. [4]

Taking L.H.S:

$$\log_b \sqrt{\frac{27}{16}} = \log_b \left(\frac{27}{16}\right)^{\frac{1}{2}}$$

$$= \log_b \left[\frac{(3)^3}{(4)^2}\right]^{\frac{1}{2}}$$

$$= \log_b \left[\frac{(3)^{\frac{3}{2}}}{4}\right]$$

$$= \log_b (3)^{\frac{3}{2}} - \log_b 4$$

$$= \frac{3}{2}\log_b 3 - \log_b 4$$

Substituting $\log_b 3 = m$ and $\log_b 4 = n$ gives,

$$\log_b \sqrt{\frac{27}{16}} = \frac{3}{2}(m) - (n)$$

$$= \frac{3m}{2} - n$$

$$= \text{R.H.S.}$$

$$\therefore \log_b \sqrt{\frac{27}{16}} = \frac{3m}{2} - n$$

Q.E.D.

(ii) Solve the equation $\log_2 m + \log_m 16 = 4$. [6]

$$\log_2 m + \log_m 16 = 4$$

$$\log_2 m + \frac{\log_2 16}{\log_2 m} = 4$$

$$\log_2 m + \frac{4}{\log_2 m} = 4$$

Let $x = \log_2 m$.

$$x + \frac{4}{x} = 4$$

$(\times\, x)$

$$x^2 + 4 = 4x$$

$$x^2 - 4x + 4 = 0$$

$$(x-2)^2 = 0$$

$$x - 2 = 0$$

$$x = 2$$

Now,

$$\log_2 m = 2$$

Converting to exponential form gives:

$$m = (2)^2$$

$$= 4$$

$\therefore$ The value of $m = 4$.

(b) Use mathematical induction to prove that $7^n - 1$ is divisible by 6 for all $n \in \mathbb{N}$. [9]

Let $P(n)$ be the statement: $7^n - 1$ is divisible by 6 for $n \in \mathbb{N}$

Looking at $P(1)$:

$7^1 - 1 = 6$ which is divisible by 6.

So, $P(n)$ is true for $n = 1$.

Assume $P(k)$ is true.

That is, $7^k - 1$ is divisible by 6

We have: $7^k - 1 = 6m$, where $m \in \mathbb{N}$

$\Rightarrow 7^k = 6m + 1$

Looking at $P(k+1)$.

That is, $7^{k+1} - 1$ is divisible by 6.

We need to prove $P(k+1)$.

$$\begin{aligned} 7^{k+1} - 1 &= 7^k \cdot 7 - 1 \\ &= (6m+1)(7) - 1 \\ &= 42m + 7 - 1 \\ &= 42m + 6 \\ &= 6(7m+1) \quad \text{which is divisible by 6} \end{aligned}$$

So, if $P(k)$ is true, $P(k+1)$ is also true.

Since $P(1)$ is true, and since $P(k+1)$ is true when $P(k)$ is true, then by the principle of mathematical induction, $P(n)$ is true for all $n$, ie.

$7^n - 1$ is divisible by 6 for all $n \in \mathbb{N}$.

Q.E.D.

(c) A binary operation $*$ is defined on $\mathbb{R}$ by $x * y = \left(x^{\frac{1}{3}} + y^{\frac{1}{3}}\right)^3$.

(i) Calculate $8 * 27$. [3]

$$x * y = \left(x^{\frac{1}{3}} + y^{\frac{1}{3}}\right)^3$$

$$8 * 27 = \left[(8)^{\frac{1}{3}} + (27)^{\frac{1}{3}}\right]^3$$

$$= \left[(2^3)^{\frac{1}{3}} + (3^3)^{\frac{1}{3}}\right]^3$$

$$= (2+3)^3$$

$$= (5)^3$$

$$= 125$$

(ii) Show that $*$ is commutative. [3]

$$x * y = \left(x^{\frac{1}{3}} + y^{\frac{1}{3}}\right)^3$$

$$= \left(y^{\frac{1}{3}} + x^{\frac{1}{3}}\right)^3$$

$$= y * x$$

Since $x * y = y * x$, then $*$ is commutative.

Q.E.D.

**Total: 25 marks**

June 2025 – Question 3 – Solution

(a) A circle has centre $C(1,3)$ and passes through the point $A(3,7)$.

(i) Determine the equation of the circle. [4]

The distance between the centre $C(1,3)$ and the point $A(3,7)$ is the radius.

$$\text{Radius} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(3-1)^2 + (7-3)^2}$$

$$= \sqrt{(2)^2 + (4)^2}$$

$$= \sqrt{4 + 16}$$

$$= \sqrt{20} \text{ units}$$

The equation of the circle is:

$$(x-a)^2 + (y-b)^2 = r^2$$

$$(x-1)^2 + (y-3)^2 = \left(\sqrt{20}\right)^2$$

$$(x-1)^2 + (y-3)^2 = 20$$

$\therefore$ The equation of the circle is: $(x-1)^2 + (y-3)^2 = 20$

(ii) Show that the equation of the tangent to the circle at the point $A$ is

$2y + x - 17 = 0.$ [4]

Points are $C(1, 3)$ and $A(3, 7)$.

Gradient of normal $= \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{7-3}{3-1}$

$= \frac{4}{2}$

$= 2$

Gradient of perpendicular line $= -1 \div 2$

$= -1 \times \frac{1}{2}$

$= -\frac{1}{2}$

Substituting $m = -\frac{1}{2}$ and point $A(3, 7)$ into $y - y_1 = m(x - x_1)$ gives:

$y - 7 = -\frac{1}{2}(x - 3)$

$y - 7 = -\frac{1}{2}x + \frac{3}{2}$

$(\times 2)$

$2y - 14 = -x + 3$

$2y + x - 14 - 3 = 0$

$2y + x - 17 = 0$

$\therefore$ The equation of the tangent to the circle is $2y + x - 17 = 0$.

Q.E.D.

(b) Determine the points of intersection of the parabola

$x^2 - 3x - 5 - y = 0$ and the line $2x + y + 3 = 0$. [7]

$x^2 - 3x - 5 - y = 0 \quad \rightarrow$ Equation 1

$2x + y + 3 = 0 \quad \rightarrow$ Equation 2

Rearranging Equation 1 gives:

$y = -2x - 3 \quad \rightarrow$ Equation 3

Substituting Equation 3 into Equation 1 gives:

$$x^2 - 3x - 5 - (-2x - 3) = 0$$

$$x^2 - 3x - 5 + 2x + 3 = 0$$

$$x^2 - x - 2 = 0$$

$$(x + 1)(x - 2) = 0$$

Either $x + 1 = 0$ or $x - 2 = 0$

$x = -1$ $\quad$ $x = 2$

When $x = -1$,

$y = -2(-1) - 3$

$= 2 - 3$

$= -1$

When $x = 2$,

$y = -2(2) - 3$

$= -4 - 3$

$= -7$

$\therefore$ The points of intersection are $(-1, -1)$ and $(2, -7)$.

(c) Two vectors are given as $\overrightarrow{OA} = \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} 2 \\ 0 \\ 5 \end{pmatrix}$.

(i) Calculate $|\overrightarrow{AB}|$. [4]

Using the triangle law,

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$$

$$= \begin{pmatrix} 2 \\ 0 \\ 5 \end{pmatrix} - \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}$$

$$= \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$$

Now,

$$|\overrightarrow{AB}| = \sqrt{x^2 + y^2 + z^2}$$

$$= \sqrt{(3)^2 + (-1)^2 + (2)^2}$$

$$= \sqrt{9 + 1 + 4}$$

$$= \sqrt{14} \text{ units}$$

(ii) Determine the angle between the vectors. [6]

$$\overrightarrow{OA} \cdot \overrightarrow{OB} = \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 0 \\ 5 \end{pmatrix}$$

$$= (-1)(2) + (1)(0) + (3)(5)$$

$$= -2 + 0 + 15$$

$$= 13$$

$$|\overrightarrow{OA}| = \sqrt{x^2 + y^2 + z^2}$$

$$= \sqrt{(-1)^2 + (1)^2 + (3)^2}$$

$$= \sqrt{1 + 1 + 9}$$

$$= \sqrt{11} \text{ units}$$

$$|\overrightarrow{OB}| = \sqrt{x^2 + y^2 + z^2}$$

$$= \sqrt{(2)^2 + (0)^2 + (5)^2}$$

$$= \sqrt{4 + 0 + 25}$$

$$= \sqrt{29} \text{ units}$$

Now,

$$\cos\theta = \frac{\overrightarrow{OA} \cdot \overrightarrow{OB}}{|\overrightarrow{OA}||\overrightarrow{OB}|}$$

$$\cos\theta = \frac{13}{\sqrt{11}\sqrt{29}}$$

$$\theta = \cos^{-1}\left(\frac{13}{\sqrt{11}\sqrt{29}}\right)$$

$$\theta = 43.3° \qquad \text{(to 1 decimal place)}$$

$\therefore$ The angle between the vectors is $\theta = 43.3°$.

**Total: 25 marks**

## June 2025 – Question 4 – Solution

(a) Show that $\frac{1-\cos 2\alpha}{\sin 2\alpha} \equiv \tan \alpha$. [5]

Taking L.H.S:

$$\frac{1-\cos 2\alpha}{\sin 2\alpha} = \frac{1-(1-2\sin^2 \alpha)}{2\sin \alpha \cos \alpha}$$

$$= \frac{1-1+2\sin^2 \alpha}{2\sin \alpha \cos \alpha}$$

$$= \frac{2\sin^2 \alpha}{2\sin \alpha \cos \alpha}$$

$$= \frac{2\sin \alpha(\sin \alpha)}{2\sin \alpha(\cos \alpha)}$$

$$= \frac{\sin \alpha}{\cos \alpha}$$

$$= \tan \alpha$$

$$= \text{R.H.S.}$$

$$\therefore \frac{1-\cos 2\alpha}{\sin 2\alpha} = \tan \alpha$$

Q.E.D.

(b) Solve the equation $\tan^2 x - \tan x = 0$ for $-\pi < x < \pi$. [5]

$$\tan^2 x - \tan x = 0$$

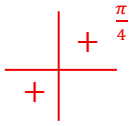
$$\tan x\,(\tan x - 1) = 0$$

Either $\tan x = 0$ or $\tan x - 1 = 0$

$x = 0$ $\tan x = 1$

$x = \frac{\pi}{4}, -\left(\pi - \frac{\pi}{4}\right)$

$x = \frac{\pi}{4}, -\frac{3\pi}{4}$




$\therefore x = -\frac{3\pi}{4}, 0, \frac{\pi}{4}$ for $-\pi < x < \pi$.

(c) Let $g(\theta) = \sqrt{3}\cos\theta - \sin\theta$.

(i) Show that $g(\theta) = 2\cos\left(\theta + \frac{\pi}{6}\right)$. [5]

Let $\sqrt{3}\cos\theta - \sin\theta = R\cos(\theta + \alpha)$

$$\sqrt{3}\cos\theta - \sin\theta = R\cos\theta\cos\alpha - R\sin\theta\sin\alpha$$

Comparing and equating coefficients gives:

$R\cos\alpha = \sqrt{3}$ → Equation 1

$R\sin\alpha = 1$ → Equation 2

Equation 2 ÷ Equation 1 gives:

$$\frac{R\sin\alpha}{R\cos\alpha} = \frac{1}{\sqrt{3}}$$

$$\tan\alpha = \frac{1}{\sqrt{3}}$$

$$\alpha = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$\alpha = \frac{\pi}{6}$$

$$R = \sqrt{\left(\sqrt{3}\right)^2 + (1)^2}$$

$$= \sqrt{3+1}$$

$$= \sqrt{4}$$

$$= 2 \text{ units}$$

$$\therefore g(\theta) = 2\cos\left(\theta + \frac{\pi}{6}\right)$$

Q.E.D.

(ii) Hence, or otherwise, determine the general solution of $g$. [4]

$g(\theta) = 2\cos\left(\theta + \frac{\pi}{6}\right)$

Let $g(\theta) = 0$.

$2\cos\left(\theta + \frac{\pi}{6}\right) = 0$

$\cos\left(\theta + \frac{\pi}{6}\right) = 0$

$PV = \frac{\pi}{2}$

$\therefore$ G.S. is $\theta + \frac{\pi}{6} = 2n\pi \pm \frac{\pi}{2}$ , $n \in \mathbb{Z}$

$\theta = 2n\pi \pm \frac{\pi}{2} - \frac{\pi}{6}$ , $n \in \mathbb{Z}$

Alternatively,

$g(\theta) = \sqrt{3}\cos\theta - \sin\theta$

Let $g(\theta) = 0$.

$\sqrt{3}\cos\theta - \sin\theta = 0$

$\sqrt{3}\cos\theta = \sin\theta$

$\sqrt{3} = \frac{\sin\theta}{\cos\theta}$

$\sqrt{3} = \tan\theta$

$\tan\theta = \sqrt{3}$

$PV = \frac{\pi}{3}$

$\therefore$ G.S. is $\theta = n\pi + \frac{\pi}{3}$ , $n \in \mathbb{Z}$

(d) Determine the Cartesian equation of a line, $l$, whose parametric equations are given as $x = -5 - 6t$ and $y = -2 - 8t$. [3]

$x = -5 - 6t \quad \rightarrow$ Equation 1

$y = -2 - 8t \quad \rightarrow$ Equation 2

Rearranging Equation 1 gives:

$x = -5 - 6t$

$x + 5 = -6t$

$\frac{x+5}{-6} = t \quad \rightarrow$ Equation 3

Substituting Equation 3 into Equation 2 gives:

$y = -2 - 8\left(\frac{x+5}{-6}\right)$

$y = -2 - \frac{8}{-6}(x + 5)$

$y = -2 + \frac{4}{3}(x + 5)$

$y = -2 + \frac{4}{3}x + \frac{20}{3}$

$y = \frac{4}{3}x + \frac{14}{3}$

Multiplying throughout by 3 gives:

$3y = 4x + 14$

$\therefore$ The Cartesian equation is: $3y = 4x + 14$

(e) A plane, $P$, passes through $A(-3,2,1)$, $B(1,1,1)$ and $C(-2,0,-1)$, and has a normal vector $n = \begin{pmatrix} 2 \\ 8 \\ -7 \end{pmatrix}$. Determine the vector equation of the plane $P$. [3]

Given the point $A(-3,2,1)$, then let $\boldsymbol{a} = \begin{pmatrix} -3 \\ 2 \\ 1 \end{pmatrix}$.

The vector equation of the plane is:

$\boldsymbol{r} \cdot \boldsymbol{n} = \boldsymbol{a} \cdot \boldsymbol{n}$

$$\boldsymbol{r} \cdot \begin{pmatrix} 2 \\ 8 \\ -7 \end{pmatrix} = \begin{pmatrix} -3 \\ 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 8 \\ -7 \end{pmatrix}$$

$$\boldsymbol{r} \cdot \begin{pmatrix} 2 \\ 8 \\ -7 \end{pmatrix} = (-3)(2) + (2)(8) + (1)(-7)$$

$$\boldsymbol{r} \cdot \begin{pmatrix} 2 \\ 8 \\ -7 \end{pmatrix} = -6 + 16 - 7$$

$$\boldsymbol{r} \cdot \begin{pmatrix} 2 \\ 8 \\ -7 \end{pmatrix} = 3$$

**Total: 25 marks**

## June 2025 – Question 5 – Solution

(a) Evaluate $\lim_{x\to 0}\frac{\sin 3x}{2x}$. [3]

$$\lim_{x\to 0}\frac{\sin 3x}{2x} = \frac{1}{2}\lim_{x\to 0}\frac{\sin 3x}{x}$$

$$= \frac{1}{2}\lim_{x\to 0}\left(\frac{\sin 3x}{x}\cdot\frac{3}{3}\right)$$

$$= \frac{3}{2}\lim_{x\to 0}\left(\frac{\sin 3x}{3x}\right)$$

$$= \frac{3}{2}(1)$$

$$= \frac{3}{2}$$

(b) Using first principles, differentiate $5x^2 + 4x - 2$ with respect to $x$. [6]

Let $f(x) = 5x^2 + 4x - 2$.

Using first principles,

$$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

$$= \lim_{h\to 0}\frac{[5(x+h)^2+4(x+h)-2]-[5x^2+4x-2]}{h}$$

$$= \lim_{h\to 0}\frac{5(x^2+2xh+h^2)+4x+4h-2-5x^2-4x+2}{h}$$

$$= \lim_{h\to 0}\frac{5x^2+10xh+5h^2+4x+4h-2-5x^2-4x+2}{h}$$

$$= \lim_{h\to 0}\frac{10xh+5h^2+4h}{h}$$

$$= \lim_{h\to 0}\frac{h(10x+5h+4)}{h}$$

$$= \lim_{h\to 0}(10x+5h+4)$$

$$= 10x + 5(0) + 4$$

$$= 10x + 4$$

(c) Given that $y = x^3 \sin^4(3x^2 - 2)$, determine $\frac{dy}{dx}$. [5]

$y = x^3 \sin^4(3x^2 - 2)$

Let $u = x^3$ and $v = \sin^4(3x^2 - 2)$

Consider $u = x^3$.

$\frac{du}{dx} = 3x^2$

Consider,

$v = \sin^4(3x^2 - 2)$

$= [\sin(3x^2 - 2)]^4$

Now,

$\frac{dv}{dx} = 4[6x\cos(3x^2 - 2)][\sin(3x^2 - 2)]^3$

$= 24x\cos(3x^2 - 2)\sin^3(3x^2 - 2)$

Using the product rule,

$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$

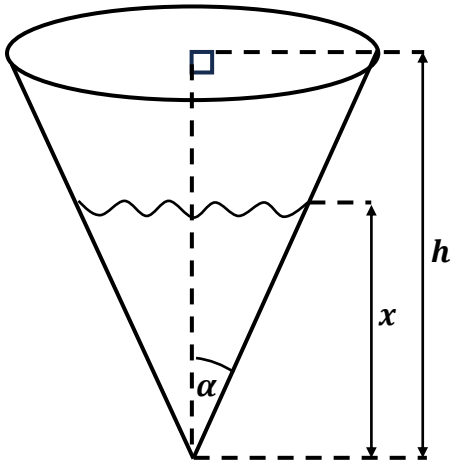
$= (x^3)[24x\cos(3x^2 - 2)\sin^3(3x^2 - 2)] + [\sin^4(3x^2 - 2)](3x^2)$

$= 24x^4\sin^3(3x^2 - 2)\cos(3x^2 - 2) + 3x^2\sin^4(3x^2 - 2)$

$\therefore \frac{dy}{dx} = 24x^4\sin^3(3x^2 - 2)\cos(3x^2 - 2) + 3x^2\sin^4(3x^2 - 2)$

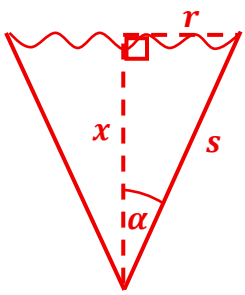
(d) Water is flowing into an empty right circular cone at a constant rate of 8 $cm^3$ per minute. The depth of the liquid in the cone at time, $t$, in minutes, is $x\ cm$.

The angle, $\alpha$, between the perpendicular height, $h$, of the cone and the slant height is $\tan^{-1}\frac{\sqrt{3}}{3}$ and the volume of the cone can be calculated as $V = \frac{1}{3}\pi r^2 h$.




(i) Show that the volume, $V\ cm^3$, of water in the cone when the depth is $x\ cm$ can be expressed as $V = \frac{1}{9}\pi x^3$. [5]

Consider the sketch below:




We are given that:

$$\alpha = \tan^{-1}\frac{\sqrt{3}}{3}$$

$$\tan\alpha = \frac{\sqrt{3}}{3}$$

Now,

$$\tan\alpha = \frac{opp}{adj}$$

$$\tan\left(\frac{\pi}{6}\right) = \frac{r}{x}$$

$$\frac{\sqrt{3}}{3} = \frac{r}{x}$$

$$3r = \sqrt{3}x$$

$$r = \frac{\sqrt{3}x}{3}$$

The volume of the water in the cone is:

$$V = \frac{1}{3}\pi r^2 h$$

$$= \frac{1}{3}\pi\left(\frac{\sqrt{3}x}{3}\right)^2(x)$$

$$= \frac{1}{3}\pi\left(\frac{3x^2}{9}\right)(x)$$

$$= \frac{1}{3}\pi\left(\frac{x^2}{3}\right)(x)$$

$$= \frac{1}{9}\pi x^3$$

$\therefore$ The volume of the water in the cone is $V = \frac{1}{9}\pi x^3$.

Q.E.D.

(ii) Calculate the rate at which the height of the water is increasing when the height of the water in the cone is 6 $cm$. [6]

We are given that $\frac{dV}{dt} = 8\ cm^3$ per minute.

$$V = \frac{1}{9}\pi x^3$$

$$\frac{dV}{dx} = \frac{1}{9}(3)\pi x^{3-1}$$

$$= \frac{1}{3}\pi x^2$$

When $x = 6$,

$$\frac{dV}{dx} = \frac{1}{3}\pi(6)^2$$

$$= \frac{1}{3}\pi \times 36$$

$$= 12\pi$$

Using chain rule,

$$\frac{dx}{dt} = \frac{dx}{dV} \times \frac{dV}{dt}$$

$$= \frac{1}{12\pi} \times 8$$

$$= \frac{2}{3\pi}\ cm \text{ per minute}$$

$\therefore$ The rate at which the height of the water is increasing is $\frac{2}{3\pi}\ cm$ per minute.

**Total: 25 marks**

June 2025 – Question 6 – Solution

(a) The mass of a population of a certain microbe triples every 4 hours and after 6 hours the mass of the population is 375 $g$. Calculate the mass of the population of microbes at time $t = 0$.

Note that $\int \frac{1}{y}\, dy = \ln y + c$. [10]

From the information given, we can deduce that the rate of increase of the mass of a population $y$ at time $t$ is directly proportional to the population $y$ at time $t$.

$\frac{dy}{dt} \propto y$

$\frac{dy}{dt} = ky$

Separating variables:

$\frac{1}{y}\, dy = k\, dt$

Integrating both sides:

$\int \frac{1}{y}\, dy = \int k\ dt$

$\ln y = kt + c$

$y = e^{kt+c}$

$y = e^{kt} \cdot e^{c}$

$A = A_0 e^{kt}$ where $A = y$ (the mass of a population at time $t$)

and $A_0 = e^{c}$ (the mass of a population at time $t = 0$)

We are given that the mass of a population triples every 4 hours.

So, we have,

$3A_0 = A_0 e^{k(4)}$

$3 = e^{4k}$

Taking ln on both sides:

$\ln 3 = \ln e^{4k}$

$\ln 3 = 4k \ln e$

$\ln 3 = 4k(1)$

$\ln 3 = 4k$

$\frac{\ln 3}{4} = k$

We are given that after 6 hours the mass of the population is 375 $g$.

$375 = A_0 e^{k(6)}$

$375 = A_0 e^{6k}$

Substituting $k = \frac{\ln 3}{4}$ gives:

$375 = A_0 e^{6\left(\frac{\ln 3}{4}\right)}$

$375 = A_0 e^{\left(\frac{3}{2}\right)\ln 3}$

$375 = A_0 e^{\ln(3)^{\frac{3}{2}}}$

$375 = A_0 (3)^{\frac{3}{2}}$

$375 = A_0 \left(\sqrt{3}\right)^3$

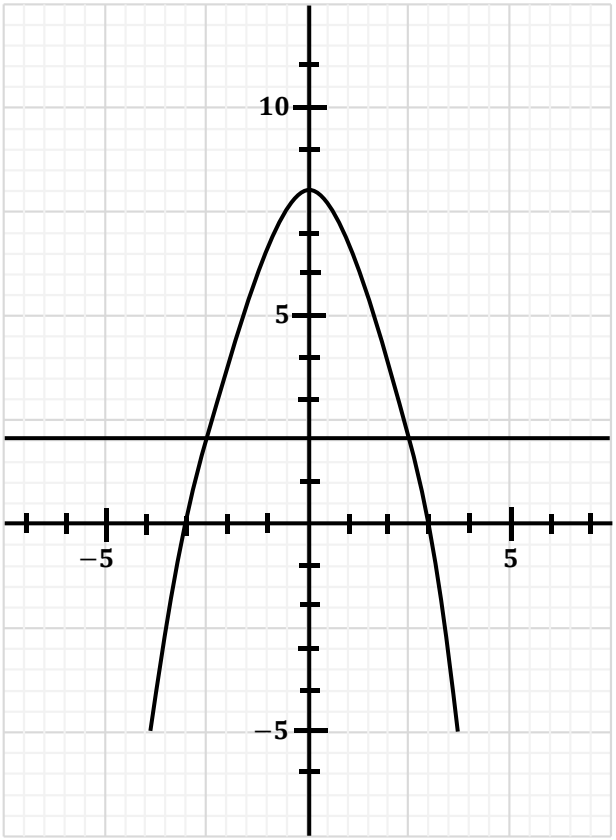
$375 = A_0 \left(3\sqrt{3}\right)$

$\frac{375}{3\sqrt{3}} = A_0$

$\frac{125}{\sqrt{3}} = A_0$

$\therefore$ The mass of the population of microbes at time $t = 0$ is $A_0 = \frac{125}{\sqrt{3}}$ $g$.

(b) The diagram below, **not drawn to scale**, shows the graphs of the curve $y = 8 - x^2$ and the line $y = 2$.




Calculate the volume of the solid of revolution generated by revolving the region bounded by the curve $y = 8 - x^2$ and the line $y = 2$ about the $x$-axis. [9]

We first need to find the $x$-values of the points of intersection.

To do, we solve the equations simultaneously.

$y = 8 - x^2$ → Equation 1

$y = 2$ → Equation 2

Equating both equations gives:

$$2 = 8 - x^2$$

$$x^2 = 8 - 2$$

$$x^2 = 6$$

$$x^2 - 6 = 0$$

$$(x + \sqrt{6})(x - \sqrt{6}) = 0$$

Either $x + \sqrt{6} = 0$ or $x - \sqrt{6} = 0$

$x = -\sqrt{6}$ $x = \sqrt{6}$

Let $V_1$ be the volume of the solid of revolution of the curve about the $x$-axis.

Let $V_2$ be the volume of the solid of revolution of the line about the $x$-axis.

The limits are $x = -\sqrt{6}$ and $x = \sqrt{6}$.

Consider the curve $y = 8 - x^2$.

$$y^2 = (8 - x^2)^2$$

$$= 64 - 16x^2 + x^4$$

Consider the line $y = 2$.

$$y^2 = (2)^2$$

$$= 4$$

Note that the volume of the solid about the $x$-axis $= \pi \int_a^b y^2 \, dx$

$$\text{Volume} = V_1 - V_2$$

$$= \pi \int_a^b (y_1)^2\, dx - \pi \int_a^b (y_2)^2\, dx$$

$$= \pi \int_{-\sqrt{6}}^{\sqrt{6}} (64 - 16x^2 + x^4)\, dx \; - \pi \int_{-\sqrt{6}}^{\sqrt{6}} (4)\, dx$$

$$= \pi \int_{-\sqrt{6}}^{\sqrt{6}} (64 - 16x^2 + x^4 - 4)\, dx$$

$$= \pi \int_{-\sqrt{6}}^{\sqrt{6}} (60 - 16x^2 + x^4)\, dx$$

$$= \pi \left[60x - \frac{16x^3}{3} + \frac{x^5}{5}\right]_{-\sqrt{6}}^{\sqrt{6}}$$

$$= \pi \left\{\left[60(\sqrt{6}) - \frac{16(\sqrt{6})^3}{3} + \frac{(\sqrt{6})^5}{5}\right] - \left[60(-\sqrt{6}) - \frac{16(-\sqrt{6})^3}{3} + \frac{(-\sqrt{6})^5}{5}\right]\right\}$$

$$= \pi \left[\left(60\sqrt{6} - 32\sqrt{6} + \frac{36}{5}\sqrt{6}\right) - \left(-60\sqrt{6} + 32\sqrt{6} - \frac{36}{5}\sqrt{6}\right)\right]$$

$$= \pi \left(60\sqrt{6} - 32\sqrt{6} + \frac{36}{5}\sqrt{6} + 60\sqrt{6} - 32\sqrt{6} + \frac{36}{5}\sqrt{6}\right)$$

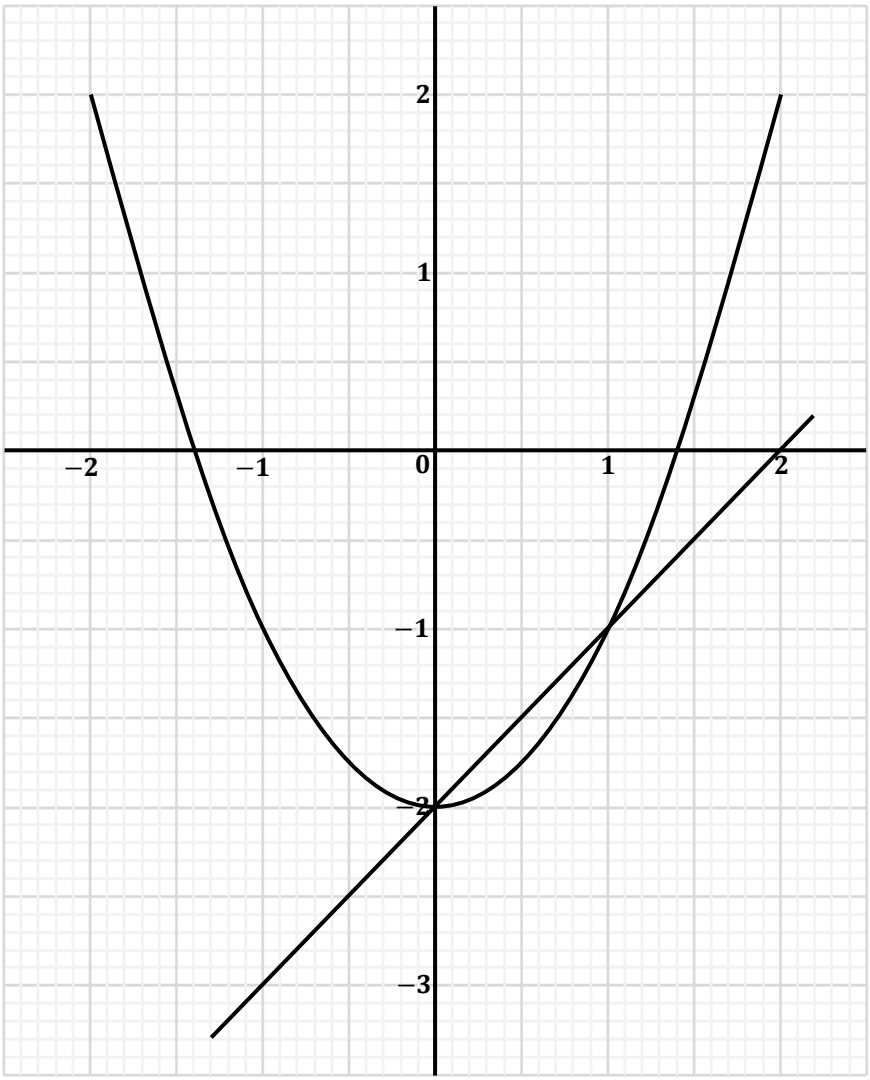
$$= \pi \left(120\sqrt{6} - 64\sqrt{6} + \frac{72}{5}\sqrt{6}\right)$$

$$= \pi \left(\frac{352}{5}\sqrt{6}\right)$$

$$= \frac{352}{5}\pi\sqrt{6} \text{ units}^3$$

$\therefore$ The volume of the solid of revolution generated is $\frac{352}{5}\pi\sqrt{6}$ units$^3$.

(c) The diagram below, **not drawn to scale**, shows the curve $y = x^2 - 2$ and the line $y = x - 2$.




Calculate the area of the region enclosed by the curve and the line. [6]

Let $y_1 = x^2 - 2$ and $y_2 = x - 2$.

From the graph, the $x$-limits are $x = 0$ and $x = 1$.

Now,

Area of the region $= \int_a^b (y_2 - y_1)\, dx$

$$= \int_0^1 [(x-2) - (x^2 - 2)]\, dx$$

$$= \int_0^1 (x - 2 - x^2 + 2)\, dx$$

$$= \int_0^1 (x - x^2)\, dx$$

$$= \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1$$

$$= \left[\frac{(1)^2}{2} - \frac{(1)^3}{3}\right] - \left[\frac{(0)^2}{2} - \frac{(0)^3}{3}\right]$$

$$= \left(\frac{1}{2} - \frac{1}{3}\right) - 0$$

$$= \frac{1}{6}\text{units}^2$$

$\therefore$ The area of the region enclosed by the curve and the line is $\frac{1}{6}\text{units}^2$.

**Total: 25 marks**

**END OF TEST**

**IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.**